# FEASIBILITY OF MAPPING TURBULENCE IN REAL-TIME USING MODE-S DATA

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## ABSTRACT

*The Met Office has identified the potential to use Mode-S data in turbulence monitoring, and particularly the use of Roll angle, to improve current forecasting methods. Mode-S is a Secondary Surveillance Radar process that allows selective interrogation of aircraft, according to the unique 24-bit address assigned to each aircraft. It is a publicly available data source and is frequently sent down via transponders and can be decoded live to retrieve individual aircraft parameters, including Roll angle and Vertical rate. Mode-S as a source of data and its application to reliable live mapping is explored. This report analyses real flights' Vertical rate and Roll angle reports and assesses their accuracy in predicting turbulence presence and intensity. The parameters as objective indicators of turbulence are considered by inputting known intensities of turbulence into an aircraft model and recording the responses. A correlation between Vertical rate and Roll angle standard deviations across 30 second intervals to turbulence intensities were shown. Mode-S data was also shown to be able to be graphically outputted within a 3 minute time scale.*

**Keywords:** Mode-S, Turbulence, Roll angle, Vertical rate, Live-mapping, ADSB.

## 1. ABBREVIATIONS

| | | | |
|---|---|---|---|
| ADSB | Automatic Dependent Surveillance-Broadcast | IFF | Identification Friend or Foe |
| AMDAR | Aircraft Meteorological Data Relay | MHR | Meteorological Hazard Report |
| BDS | Comm-B Data Selector | Mode-S | Mode-Selective |
| BDS4,0 | Vertical intention report | MRAR | Meteorological Routine Air Report |
| BDS4,4 | Meteorological Routine Air Report | NaN | Not a Number |
| BDS4,5 | Meteorological Hazard Report | NWP | Numerical Weather Prediction |
| BDS5,0 | Track and turn report | RASD | Roll angle Standard Deviation |
| BDS6,0 | Heading and speed report | RCAM | Research Civil Aircraft Model |
| CAT | Clear Air Turbulence | SIGWX | Significant Weather Chart |
| EHS | Enhanced Surveillance | SSR | Secondary Surveillance Radar |
| ELS | Elementary Surveillance | TCAC | Tropical Cyclone Advisory Centres |
| FAA | Federal Aviation Administration | VAAC | Volcanic Ash Advisory Centres |
| GARTEUR | Group for Aeronautical Research and Technology in Europe | VRSD | Vertical rate Standard Deviation |
| ICAO | International Civil Aviation Organisation | WAFC | World Area Forecast Centres |

## 2. INTRODUCTION

Turbulence is a phenomenon where strong sudden movements occur within air or water [1]. Commonly, there are four fundamental causes of turbulence for aircraft, namely shear, thermal, mechanical, and aerodynamic [1]. Shear turbulence is the result of wind directions changing drastically within a short range. Thermal causes of turbulence refer to the formation of vertical air flows due to an increase in surface temperature. Mechanical causes of turbulence are often from the passage of strong winds over irregular terrain or obstacles and aerodynamic causes of turbulence are due to wing tip trailing vortices

generated by nearby aircraft [1]. CAT is most commonly associated with shear turbulence but can be caused by any type of turbulence.

Turbulence is an important factor to consider in aviation, as it can affect both the airframe and the ride comfort of the passengers, and consequently can have a significant impact on the safety of a flight. Weather is the second most common reason for flight delays, behind 'Capacity and Staffing', so providing pilots with accurate up-to-date meteorological information allows them to improve passenger's safety by making informed decisions on flying routes [2]. The Met Office UK is one of two World Area Forecast Centres (WAFCs) responsible for providing forecasting of weather hazards and turbulence to the aviation community [2]. The potential improvement of the accuracy of turbulence detection and mapping was the focus of this project, including a frequency of output within 3 minutes, to potentially support air traffic management live (3 minutes was the defined time scale to be 'live' data). Mode-S, where the S stands for Selective [3], is the name of a process that uses airborne transponders to provide altitude, aircraft, and identification data [4].

Figure 1 shows how Mode-S data is sent and received [5]. Mode-S is transmitted by secondary surveillance radar (SSR), which uses a radio frequency of 1030 MHz to actively interrogate the aircraft. The information then transmitted by the aircraft transponder is received using the 1090 MHz radio frequency [4]. Its origins come from wartime systems known as 'Identification Friend or Foe' (IFF) devices [3]. Mode A (Mode 3 in IFF) provides the aircraft's identity, while Mode C encodes the aircraft's altitude information, Mode A and Mode C respectively allowing the SSR to continuously interrogate and obtain this data. Both these modes are important components of air traffic control and air space management. Mode-S data was introduced later than Mode A and C, and provides additional meteorological data [3]. This report reviews the existing Mode-S research and its suitability in indicating levels of turbulence.

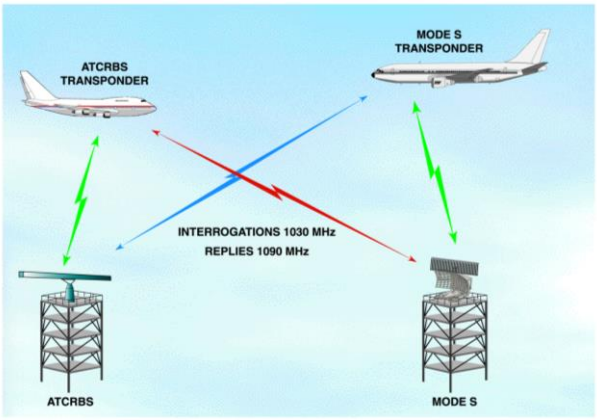



Figure 1: Diagram of Mode-S transponder setup [5]

## 3. AIMS AND OBJECTIVES

**Project Aims**: Investigation into the feasibility of using Mode-S data to achieve novel live geographical mapping of turbulence.

**Objectives:**

Initial research

- Read literature on past research.

Turbulence modelling

- Understand equations used to model turbulence.
- Create non-linearised equations of motion for aircraft dynamics, including turbulence inputs.
- Create a 6 degree of freedom model of an aircraft, including turbulence inputs.
- Investigate aircrafts response to different turbulence intensities.

Mode-S data use

- Understand transmitted report style.
- Decode data into BDS reports and parameters.
- Investigate the live calculating feasibility.
- ADSB considerations.

# 4. BACKGROUND

## 4.1 Current reporting of Mode-S

A transponder is a radio transmitter that replies to radar interrogations. For Mode-S, the transponder setup is detailed in *The Story of Mode-S* report [6]. Mode-S is different from earlier data transmission systems; previously all interrogations were general broadcasts, meaning any transponder that received an interrogation would respond. As air traffic increased, congestion problems arose and the need for discrete addressing where each transponder generates only one response became apparent. Alongside this, several air traffic accidents and the Vietnam War led to public outcry and pressure to develop better air traffic control systems, and so Mode-S came to fruition [6].

Mode-S data is split into different Comm-B Data Selector (BDS) reports [3]. Mode-S Enhanced Surveillance (EHS) data is automatically sent down as 3 reports, listed in Table 1. The reports consist of different parameters and data.

Table 1: Mode-S EHS data [3]

| Report | Parameters | Units |
|---|---|---|
| BDS4,0 | MCP/FCU selected altitude | Feet |
| | FMS Selected Altitude | Feet |
| | Barometric pressure | Millibar |
| BDS5,0 | Roll angle | Degrees |
| | True Track Angle | Degrees |
| | Ground speed | Knots |
| | Track angle rate | Degrees/second |
| | True airspeed | Knots |
| BDS6,0 | Magnetic heading | Degrees |
| | Indicated airspeed | Knots |
| | Mach number | - |
| | Barometric altitude rate | Feet/minute |
| | Inertial vertical speed | Feet/minute |

The current legal requirements of Mode-S EHS transponder equipage for flights are detailed in SKYbrary's Transponder Requirements for Access to EU Airspace webpage [7]. All fixed-wing transport-type aircraft operating in Europe with a maximum certified take-off mass exceeding 5700 kg, or having a maximum cruising true airspeed capability greater than 250 knots, are required to carry and operate Mode-S Level 2s transponder(s) with Mode-S ELS, Mode-S EHS and ADSB 1090MHz Extended Squitter (ES) capabilities by 7 June 2020 [7]. This means that all commercial aircraft flying in and out of Europe as of June 2020 have Mode-S EHS transponders.

## 4.2 Current reporting of Turbulence

Turbulence is currently mapped by a Significant Weather (SIGWX) Chart, which is produced by hand by the Met Office's WAFC meteorologists, and issued every six hours (at 0, 6, 12, 18 GMT). WAFC SIGWX forecasts are 24-hour forecasts; they represent the expected weather 24 hours after the time of observation/analysis [8]. The SIGWX charts are an essential tool for pilots, enabling them to get a broad overview of the weather relevant to their flight. The chart identifies the location of the jet streams (areas where the wind speed exceeds 80 knots), areas of potentially hazardous weather, cumulonimbus clouds, turbulence, erupting volcanoes, and tropical cyclones.

The mapped forecast areas of turbulence are drawn on the SIGWX charts by meteorologists. SIGWX forecasts only indicate regions of moderate and/or severe Clear Air Turbulence (CAT) [8]. Clear Air Turbulence is usually found above 15,000 feet, and is not associated with cumuliform clouds, including thunderstorms. It is not restricted to cloud-free air, and is the most common cause for commercial aircraft incidents in the cruise phase [9].

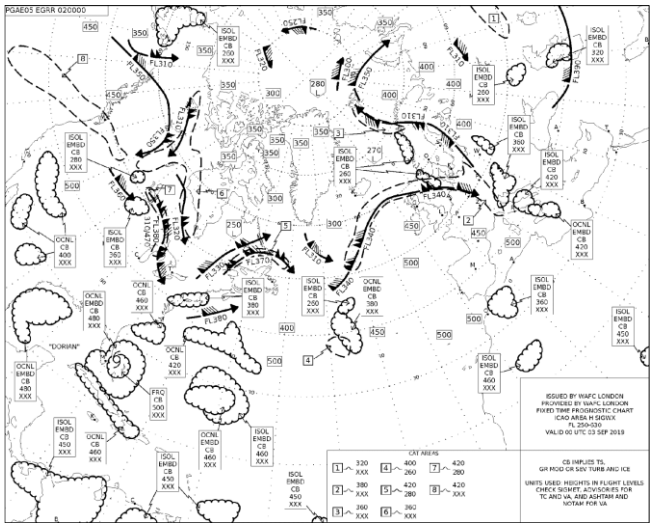


Figure 2: A SIGWX chart [8]

The atmospheric region most susceptible to CAT is the high troposphere, at aircraft cruising altitudes of around 7,000–12,000 m (23,000–39,000 ft), as it meets the tropopause [10]. Climate change is expected to strengthen the vertical wind shears at these altitudes, which would increase the prevalence of the shear instabilities that generate CAT. It is predicted that the presence of CAT will increase by 59% (Light), 94% (Moderate), and 149% (Severe) at the time of $CO_2$ doubling [11]. It will have taken approximately 200 years (from 1850 to 2050) for the first doubling of atmospheric $CO_2$ from 280 to 560 ppmv (parts per million volume), but it is predicted that the levels will double again to 1120ppmv in only 70 years or so due to the non-linear nature of greenhouse gas formation [11].

The derived field of turbulence is based on the Numerical Weather Prediction (NWP) model output which uses the Ellrod index [12]. The NWP is a form of weather model, where the input data comes from current weather observations from weather satellite imagery and radiosondes (instruments carried by balloon or other means to various levels of the atmosphere and transmitting measurements by radio). The Ellrod index is an objective technique for forecasting Clear Air Turbulence [12]. It is calculated from the numerical model output using the product of horizontal deformation, plus atmospheric convergence and vertical wind shear derived from forecast winds. The index is very useful for determining the presence of turbulence, but it has limited ability to determine the intensity of turbulence [12].

Mirza *et al* [13], explain that the Aircraft Meteorological Data Relay program (AMDAR) reports the horizontal wind and ambient temperature obtained from the aircraft's flight management system. This is useful because these parameters can be used to predict wind shear and turbulence. Meteorological Routine Air Report (MRAR) is another name for the BDS 4,4 report, and Meteorological Hazard Report (MHR) for the BDS4,5 report, which are two BDS reports that can be manually sent down to transponders by the pilot. In the *Validation of Mode-S Meteorological Routine Air Report aircraft observations* paper, it was shown that the overall quality of the Mode-S BDS4,4 report is very similar to the quality of AMDAR, and that Mode-S wind and temperature are of sufficient accuracy [14]. Mode-S BDS4,4 offers the equivalent meteorological information to that transmitted through AMDAR, therefore making Mode-S very promising for data assimilation.

## 5. PREVIOUS RESEARCH

### 5.1 Decoding Mode-S data

Sun's *The 1090 Megahertz Riddle* [15] is named as such, because Mode A/C, Mode-S, and ADSB data are all being transmitted by the aircraft across the same 1090 MHz frequency. It is a manual for decoding both Mode-S and ADSB data, from the hexadecimal binary code forms they are transmitted in. It gives an overview of the hardware and software needed to receive Mode-S signals. Similarly, de Hann et al.

[16] explain the decoding process, as well as the comparison of different parameter observations from ADSB EHS and Mode-S EHS. It was concluded that at the time there was less quality control that could be done with Mode-S EHS compared to ADSB EHS. As ADSB has since stopped recording certain parameters that are still recorded by Mode-S EHS, and there has been a larger adoption of Mode-S in Europe, it is becoming a more reliable and useful source of data [7].

### 5.2 Met Office observations

The Met Office's Observations R&D department determined an initial strong correlation between Roll angle and vertical velocity variations of a specially instrumented British Aerospace 146 aircraft flown through convective clouds [17], as shown in Figure 3. The BAE146 [18] is a small regional airliner aircraft, with a short to medium-range and a maximum take-off weight of 42,184 kg. The aircraft features 4 engines and is listed within the wake turbulence category 'Medium'. The Roll angle showed variations of between -1° and +2 °s.

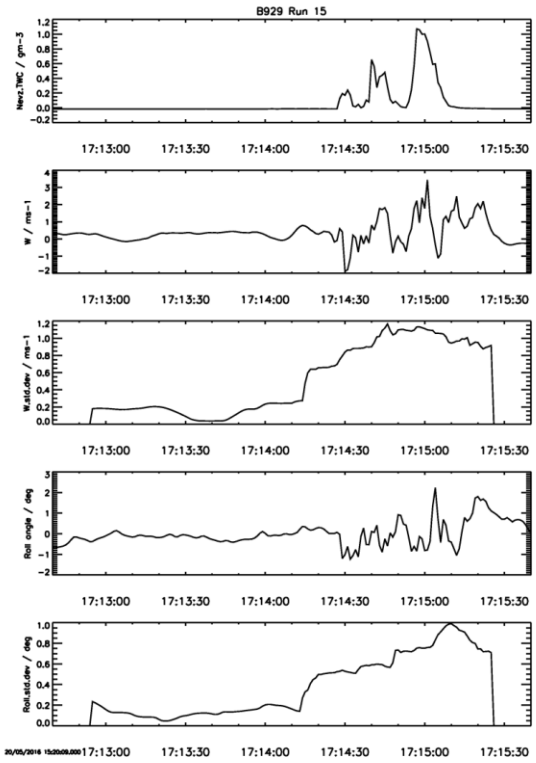



Figure 3: Met Office detection of turbulence with standard deviations [17]

Roll angle was considered to be used as a parameter to indicate turbulence. It was initially established to be potentially suitable as:

- It is a single parameter with variations reflecting a departure from 'straight and level' flight.
- It reflects air motion on similar scales to the aircraft.
- Almost every aircraft in UK airspace reports in real-time.
- It is reported with a precision of 0.17° which gives an adequate resolution.

However, possible problems using Roll angle were determined to be that it:

- Is not directly related to meteorological parameters (Eddy dissipation rates[1] or vertical acceleration) or turbulence categorisation.
- There is an unknown relationship between the Roll angle response and size of aircraft, and autopilot function/settings.
- Needs to filter out aircraft control inputs (similar to the filtering required for the derivation of horizontal winds).

For these reasons, using variations in Roll angle to indicate turbulence were assigned to be researched.

[1] Eddy Dissipation Rate is an objective, aircraft-independent, universal measure of turbulence based on the rate at which energy dissipates in the atmosphere. It is a measure of the turbulent state of the atmosphere.

### 5.3 BDS4,5 (MRAR) reporting Turbulence

Turbulence, according to the Federal Aviation Administration (FAA) in the *Aeronautical Information Manual* [19], is categorised into 4 different categories, with the BDS4,5 reporting matching these categories.

Table 2: Turbulence categorisations [19]

| Intensity | Description | BDS4,5 code |
|---|---|---|
| NIL | No turbulence | 0 |
| Light | Small changes in altitude | 1 |
| Moderate | 0.5 to 1.0 g accelerations, seat belt strain | 2 |
| Severe | Accelerations >1.0 g, temporary loss of control | 3 |
| Extreme | Total loss of control | N/A |

The turbulence reported in the BDS4,5 is still of a subjective nature and is based on the pilot's own categorisation of the intensity of the turbulence flown through. This would be considered a verification and indication of turbulence present, however, as the pilots manually have to send these down, it is currently extremely uncommon for pilots to do this. Of over 100 hundred flights BDS4,5 reports viewed, only 2 were populated with a single line of data. It has been predicted that a pilot's lack of awareness about these reports, and the reports being optional to send prevents frequent sending down. In the paper *Mode-S Transponder Comm-B Capabilities in Current Operational Aircraft* [20], it was shown that only 3.6% (1811 out of 50000) aircraft could transmit the MRAR in 2019. This additionally could be a reason as to why there are very low levels of MRAR reporting across Europe.

ICAO's *Meteorological Service for International Air Navigation* [21] manual specifies that:

- Special observations shall be made by all aircraft whenever the following conditions are encountered or observed: moderate or severe turbulence.
- Aircraft not equipped with air-ground data link shall be exempted from making routine aircraft observations.

Only 3.6% of aircraft being capable of transmitting the required moderate or severe turbulence observations further gives understanding to why these reports are infrequently sent, even when turbulence is known to be present [20]. As more aircraft become MRAR compliant, the reporting capabilities and frequency will increase and can be used alongside Mode-S EHS and other methods to provide turbulent data live.

### 5.4 Turbulence Detection using Mode-S EHS

Sun and Olive [1], determined a method to detect turbulence using Mode-S BDS5,0 reports. They used parameters decoded from Mode-S BDS5,0 report: time, and two different sources for vertical speed: a barometric raw unfiltered measure and a baro-inertial measure computed and filtered by inertial systems.

As vertical speed cannot be directly measured by sensors, a first raw estimation can be made by measuring the derivative of barometric altitude in the air data system. The barometric measurement is much noisier as it is provided unfiltered, and when compared to the inertial measurement where the noise is filtered, the distinction becomes amplified in the measured results. Turbulence is a common source of noise in the barometric measure of altitude. The differences in variance of the barometric and inertial measurements were analysed and compared. The VRSD (Vertical rate standard deviation) plotted against time was shown to be a good indicator of turbulence occurring. Pilot turbulence observations are shown to correspond well to the turbulence detected from the barometric measurement.

To verify turbulence was present on a flight, Sun and Olive themselves took multiple flights and recorded the start and end time or turbulence, speculated intensity of turbulence and when there was a pilot response to it such as the seatbelt lights being turned on.

Sun and Olive set a recommended turbulence threshold for a systematic approach on large-scale data:

$$\Delta\sigma_{V_s} \geq \overline{\Delta\sigma_{V_s}} + 1.2 \cdot \sigma(\Delta\sigma_{V_s}) \quad (1)$$

Where $\overline{\Delta\sigma_{V_s}}$ is the average standard deviation difference between barometric and inertial Vertical rate measurements during the flight. $\sigma(\Delta\sigma_{V_s})$ is the standard deviation of $\Delta\sigma_{V_s}$ over the entire flight. This equation can be interpreted as a value of $\Delta\sigma_{V_s}$. This method was recreated with Mode-S data reports from several known turbulent flights, and with modelled Dryden turbulence of various known intensities and the turbulence threshold considered.

## 6. METHODOLOGY

### 6.1 Building a model with control inputs

The Group for Aeronautical Research and Technology in Europe (GARTEUR) developed the Research Civil Aircraft Model (RCAM) [22], and produced a report that details the dimensions and parameters of an aircraft model with a mass of 120,000 kg. The Boeing 757 has a Maximum Take-Off weight of 115,66 0kg, making it a similar-sized aircraft to the RCAM, and most commonly used for long haul flights [23]. An aircraft of 120,000kg falls into the 'Medium' sized aircraft type of the ICAO Wake Turbulence Category [24]. This means for the RCAM aircraft, there is a minimum wake turbulence separation of 2 minutes, at the time aircraft are airborne. The Boeing 757 is commonly considered to be in the 'Heavy' category for the purpose of assessing wake turbulence generated by this aircraft.

The RCAM specification was used to build an RCAM Simulink and MATLAB model, with five control inputs of aileron, stabiliser, rudder, and two throttles. The model is a 6 degrees of freedom, 12-state full non-linear flight dynamic model. The 12-state variables outputs are 3 body axis translational velocity components, 3 body axis rotational velocity components, 3 Euler angles between body axis and earth axis and 3 velocities relative to the earth axis. The model was converted from a 12-state to 9-state full non-linear flight dynamic model, with the three velocities relative to the earth axis (longitude, latitude and altitude) not being included, because their response to turbulence was insignificant.

Table 3: Model state variables

| Output | Body axis | Symbol |
|---|---|---|
| Velocity | X axis | $u$ |
| Velocity | Y axis | $v$ |
| Velocity | Z axis | $w$ |
| Angular rate | X axis | $p$ |
| Angular rate | Y axis | $q$ |
| Angular rate | Z axis | $r$ |
| Bank Euler angle | X axis | $\varphi$ |
| Pitch Euler angle | Y axis | $\theta$ |
| Yaw Euler angle | Z axis | $\psi$ |

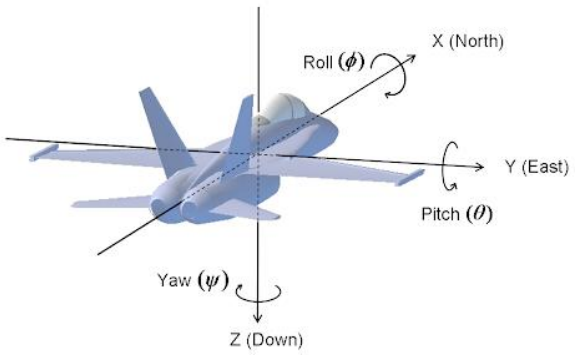



Figure 4: Model state variables representation [25]

The RCAM simulations were used to find the trim condition, by considering different elevator and thrust angles. The starting conditions for the input were determined by trimming the RCAM model with no turbulence inputs, with limits on the control inputs defined by the RCAM document included. Appendix 1 shows the trimmed RCAM model when subjected to no turbulence, and its return to its trimmed state.

Appendix 2 shows the RCAM model when subjected to severe turbulence. The Vertical rate was trimmed not at zero, but at -22.5 m/s, in order to trim the forward velocity at 237 m/s (0.8 Mach), which is the speed aircraft commonly fly at in the cruise phase. The RCAM model is built to represent an aircraft at a lower altitude, but to match the cruise phase of the real flight data, the altitude and air density of the RCAM model were adjusted to run at cruise conditions.

Table 4: Initial conditions for trimmed RCAM model

| Initial condition | Value |
|---|---|
| Forward velocity (m/s) | 242 |
| Vertical rate (m/s) | -22.5 |
| Pitch Euler angle (°) | -5 |
| Elevator (°) | -4.5 |
| Thrust (kN) | 365 |

### 6.1.1 Adding Dryden turbulence to the model

The Dryden wind turbulence model is a mathematical model of continuous gusts accepted for use by the United States Department of Defence in certain aircraft design and simulation applications [26]. The model treats the linear and angular velocity components of continuous gusts as spatially varying stochastic processes and specifies each component's power spectral density.
The Dryden Wind Turbulence Model (Continuous) block is part of the MATLAB/Simulink Aerospace Blockset library [27]. It uses the Dryden spectral representation to add turbulence to the aerospace model by passing band-limited white noise through appropriate forming filters. This block implements the mathematical representation of the MIL-F-8785C specification. The selected continuous representation of Dryden velocity spectra uses positive vertical (+q) and lateral (+r) angular rates spectra. The level of turbulence can be dictated by the Dryden turbulence block. The blocks' turbulence intensities are detailed in Table 4:

Table 5: Probability of exceedance of high-altitude intensity [27]

| Intensity | Probability |
|---|---|
| Light | $10^{-2}$ |
| Moderate | $10^{-3}$ |
| Severe | $10^{-5}$ |

The block was integrated into the RCAM model to create a simulation of a turbulent flight, with a specified intensity of turbulence. As shown in Figure 5, the outputted state variables were looped through the Dryden turbulence block, and then inputted back into the RCAM model, to create a continuous simulation of a turbulent flight. The detailed Simulink model is shown in Appendix 3.

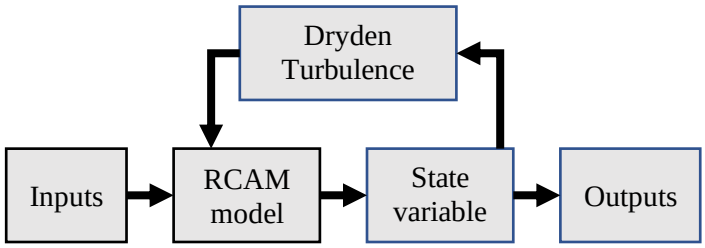



Figure 5: RCAM model with Dryden turbulence block

### 6.2 Mode-S data collection

The OpenSky Network is an independent association that was founded in 2015 to guarantee the continuous development of the network towards a completely open air traffic control sensor network with worldwide coverage. The OpenSky Network provides detailed live aircraft information over the publicly accessible 1090 MHz radio frequency channel, and all collected raw data is archived in a large

historical database for use by researchers from different areas to analyse and improve air traffic control technologies and processes [28].

The Mode-S data was accessed by interrogating the OpenSky Impala Historical database [29]. The database was populated with Mode-S data from January 2020 onwards to the present time. Due to the 2020 Coronavirus pandemic, there were significantly fewer flights across the world and Europe in 2020 and 2021 compared to before 2020. Governments across the world restricted passengers flying for emergency and permitted reasons only, in order to prevent the spread of the Coronavirus. These restrictions prevented first-hand data collection of the presence of turbulence on flights. To verify that turbulence was present, recounts of passenger's onboard flights since January 2020 were taken. This is how the flights to be analysed were selected. A disadvantage of this method against Sun and Olive's first-person data recording also means that no data on the intensity of turbulence or exact timings was present.

A Python wrapper called 'pyOpenSky' [30] was used to query the OpenSky Impala Historical database with Mode-S data. This then outputted raw Mode-S 28 digit hexadecimal codes and Sun's 'PyModeS' decoder [31] was used to assist in decoding the unique Mode-S codes into the BDS reports and parameters contained within them. The input data needed to query the database included the publicly available start time and date of the flight in the GMT time zone, the end time and date, the ADSB Hexcode (also known as the ICAO24 code) in its hexadecimal form, and the Callsign of the flight.
Code was written to link the data into arrays, and these arrays were written to Excel spreadsheets, with these Excel spreadsheets then working as an interface between the Python Mode-S outputs and inputting them into MATLAB to analyse and plot the flights' parameters. Excel was chosen as the interface, for its ease of viewing and compatibility with MATLAB, as well as MATLAB's ability to import Excel arrays with good time complexity (meaning the time a computer takes to run an algorithm is short). It is known that the Met Office's current method of processing Mode-S data involves querying it using an SQL wrapper and processing it using C++, which as a time-efficient computing language makes it even more suitable to process live data at an operational rate.

The outputted data was then formatted to suitable form, outputted to multiple Excels spreadsheets, split into ADSB, BDS4,4, BDS4,5, BDS6,0, BDS5,0 reports. One master copy was deemed unsuitable because of the time taken by the program to query such a large amount of data and write to Excel. An Excel spreadsheet containing the Time, Altitude, Speed and Rate of Climb from the ADSB report was also created. ADSB was chosen as the data sources for these parameters because of its consistent and reliable data reporting.

This report only considers the turbulence that occurs in the cruise phase of flight. The start and end times of the cruise phase were established using the library 'FlightPhase' from the Python project 'OpenAp' [32] was used to determine the phase of flight by analysing Altitude, Forward velocity, and Rate of Climb. The Cruise phase start and end times were outputted to MATLAB, where the start and end time of the cruise phase was determined to the precision of 1 second. The graphical representation of the different phases of flight is shown in Appendix 4.

The BDS5,0 (Roll angle data) and BDS6,0 (Vertical rate data) report at inconsistent time intervals. For each of these BDS reports, a MATLAB code was implemented to select the exact or closest values of time to the 'FlightPhase' defined start and time of the cruise phase. For the start time, if an exact match was not present, the next subsequent time value was taken. For the end time, the closest preceding value was taken. This meant that both the start and end time values taken from the BDS reports respectively, were within the 'FlightPhase' defined timings of the cruise phase.

### 6.2.1 False entries

It was reported by the Met Office, that they had found common signifiers of what had proved to be a false entry in the Mode-S data. These signifiers were applied to the code, to remove what is likely to

be a false entry before outputting the correct Mode-S reports.

The signifiers for BDS6,0 were that the entry is likely to be false if:

- The absolute difference between the inertial and barometric Vertical rate is higher than 2000 ft/minute or the Mach number is greater than.
- The Mach number is 0.776 and the rounded down absolute heading is 178.
- The indicated airspeed is not between 100 and 550 kts.

For BDS5,0 they were if:

- The true air speed is between 100 and 550 kts.
- The difference between the reported ground speed and the calculated ground speed from the change in GPS position is more than 10 kts.

The nature of these signifiers, for example with a Mach number of 0.776 and a rounded down absolute heading of 178, show that the Mode-S reports can be unreliable. It is difficult to know exactly just from the data whether these parameters are falsely reported or not, meaning using the current metrics, correct data could be removed, or false data is included and analysed in the data set. It is currently unknown how frequently false Mode-S data entries occur, or which data entries are false. While these reports reduce the accuracy of the data, comparing the data including and not including the predicted false reports, the trends shown are very similar suggesting the overall trends are accurate.

### 6.2.2 Reporting frequency

ADSB data is reported reliably every second, with no dropped entries. This is a different reporting frequency to Mode-S EHS data, where the BDS5,0 report (containing Roll angle data) and BDS6,0 report (containing Vertical rate data) came through at different frequencies. Appendix 5 shows the data points for 5 flights considered and their reporting frequency analysed, to determine the average reporting frequency of different reports.

Table 6: Reporting frequency of report (seconds)

| ADSB | BDS5,0 | BDS6,0 |
|---|---|---|
| 1.00 | 1.50 | 0.70 |

It is important to note, that the BDS5,0 and BDS6,0 reports are not outputted at a uniform frequency, with the reports being outputted at random times, to the precision of 0.001 seconds. This means that there can be prolonged periods of time with very few reports, and then some with very frequent reporting. It is also the case that some of these periods of time with no reports can surpass 30 seconds, meaning there is no data to analyse standard deviations from or know what the various parameters are.

## 7. RESULTS AND ANALYSIS

### 7.1 Real flight data

### 7.1.1 Full flight path

The flight analysed in this report was a Ryanair flight from Stanstead to Salzburg, 12th October 2020, with known turbulence occurring. The aircraft was a Boeing 738, which has a maximum take-off weight of 70530 kg, wake turbulence category of ‘Medium’ and it cruises at a Mach number of 0.785 [33].

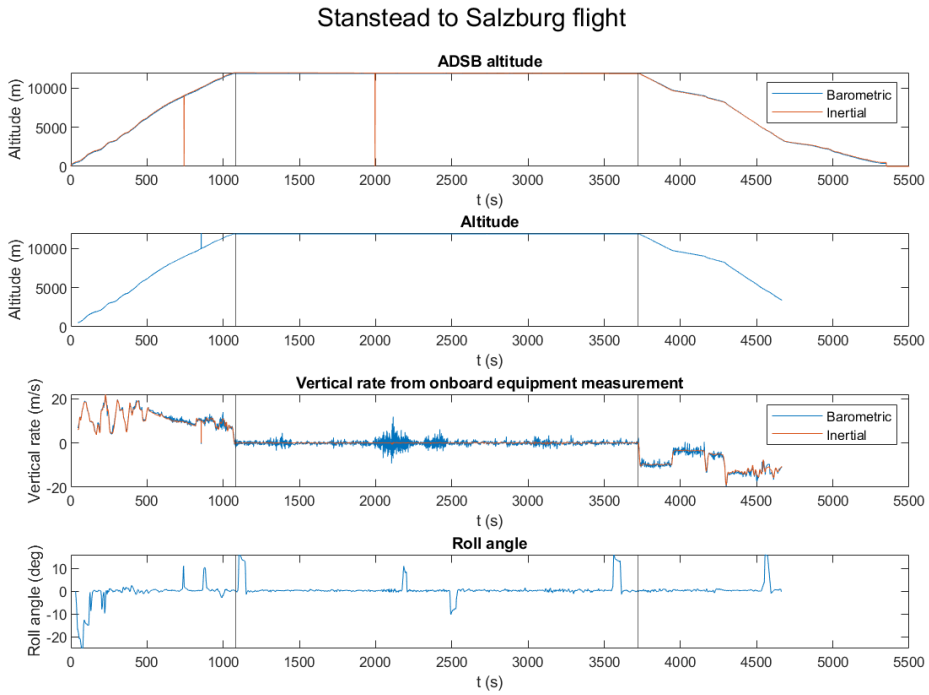



Figure 6: Stanstead to Salzburg Altitude, Vertical rate, and Roll angle

Figure 6 shows the ADSB Altitude, BDS6,0 Altitude, Vertical rate, and Roll angle during the entire flight path. Between 0 and 1100 seconds, the aircraft is in the climb phase, with the altitude increasing steadily in both graphs, and the Vertical rate being above 0, meaning the Vertical rate is increasing. The ADSB altitude is slightly different to the BDS6,0 altitude, especially in the BDS6,0 data dropping out at 4700 seconds. This shows that the ADSB data is much more reliable and hence was used for determining the start and end time of the cruise phase. The Roll angle between 0 and 200 seconds changes significantly, as the aircraft is turning onto its flight path. This is expected, as the aircraft at take-off needs to change its flight path significantly from the runway's orientation, if the destination is not in line with the runway. The cruise phase has a stable altitude and a Vertical rate of change equal to 0 in the inertial Vertical rate measurement, but around 2000 seconds the barometric altitude becomes very noisy.

ADSB altitude graph shows two dropped points in the altitude, which in ADSB data are uncommon, however in Mode-S BDS5,0 and BDS6,0 report, the altitude frequently (50% of the time) drops out and returns 'Not a Number' (NaN), which makes the Mode-S BDS6,0 much less reliable, but once all NaN's are removed, it presents a clear altitude flight path. The false BDS5,0 and BDS6,0 altitude reports do not appear to correlate to false reporting other parameters at the same time within the same entry.

### 7.1.2 Cruise phase Vertical rate

Figure 7 shows the adoption of Sun and Olive's method of comparing barometric and inertial Vertical rate measurements to detect turbulence. The 'Vertical rate from onboard equipment measurement' graph shows the raw outputs of the barometric and inertial measurements during the cruise phase of the flight, where it is clear that there is noise present at 2100 seconds.
The 'Vertical rate standard deviation across 30 second intervals' graph takes the barometric and inertial measurements individually and plots the standard deviation of points within 30 second intervals. 30 seconds was chosen interval from the Met Office's indication that 30 second intervals indicate the presence of turbulence. Applying the $\Delta\sigma_{V_s}$ threshold defined in Equation 1, it is suggested that turbulence occurs at 2100 seconds.

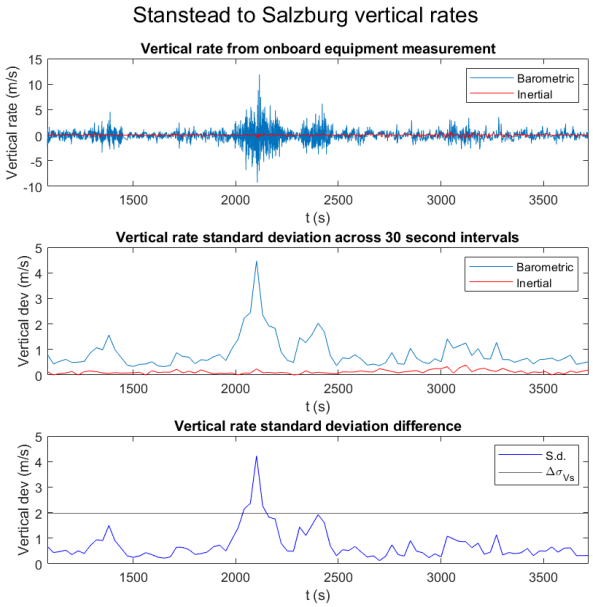



Figure 7: Stanstead to Salzburg Vertical rate in the cruise phase with 10 seconds standard deviation intervals

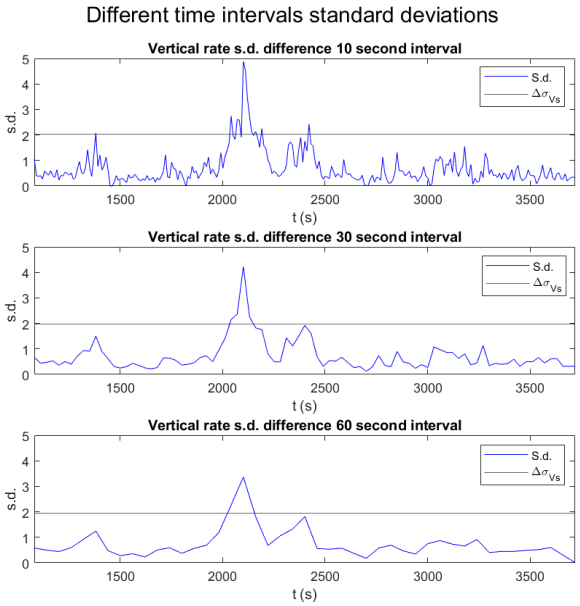



Figure 8: Standard deviation differences taken across 10, 30 and 60 second intervals.

Figure 8 shows the standard deviation difference taken across different time intervals of 10, 30 and 60 seconds. This was to discern whether the time interval impacts the detection of turbulence. The 'Vertical rate standard deviation across 10 second intervals' graph shows that a smaller interval, with the same $\Delta\sigma_{V_s}$ threshold defined in Equation 1, does indicate that more turbulence is present, compared to 30 second intervals, and again compared to 60 second intervals. The standard deviation difference crosses the $\Delta\sigma_{V_s}$ threshold 5 times, compared to the 30 and 60 second intervals where it crosses once. The difference between the $\Delta\sigma_{V_s}$ threshold across the 3 different simulations is negligible, so it can be considered constant. This indicates that different time intervals can affect the detection of turbulence using a static turbulence threshold. More research is required to establish what time interval for standard deviation to be taken at most accurately detects turbulence.

### 7.1.3 Cruise phase Roll and Track angles

Figure 9 shows the Roll angle, True Track Angle, Roll angle rate, Track angle rate and RASD (Roll angle standard deviation) of the aircraft during the cruise phase. Comparing the Roll angle and Track angle graphs, the biggest peaks in the Roll angle coincide with Track angle change. This is to be expected, as to adjust the flight path and change the Track angle, the aircraft Roll angle is adjusted which in turn leads to a Track angle change and a different course of the flight path.

The 2100 second mark shows some disturbance in the Roll angle and Roll angle rate, before the 2200 second mark where the aircraft has changes Track angle and in turn the course of the flight path. It can be speculated that as the noise in Vertical rate at 2100 seconds, that there was turbulence at this point. The pilot could have chosen to change the flight path in order to avoid a turbulent patch of air, but without knowing the pilot's control inputs, it is not possible to confirm this.

The RASD is highest at the same points where the Track angle also changes, otherwise is very small and no clear conclusion can be drawn to suggest it shows turbulence during the periods of time when the Track angle is not changing, and hence the Track angle rate is approximately 0.

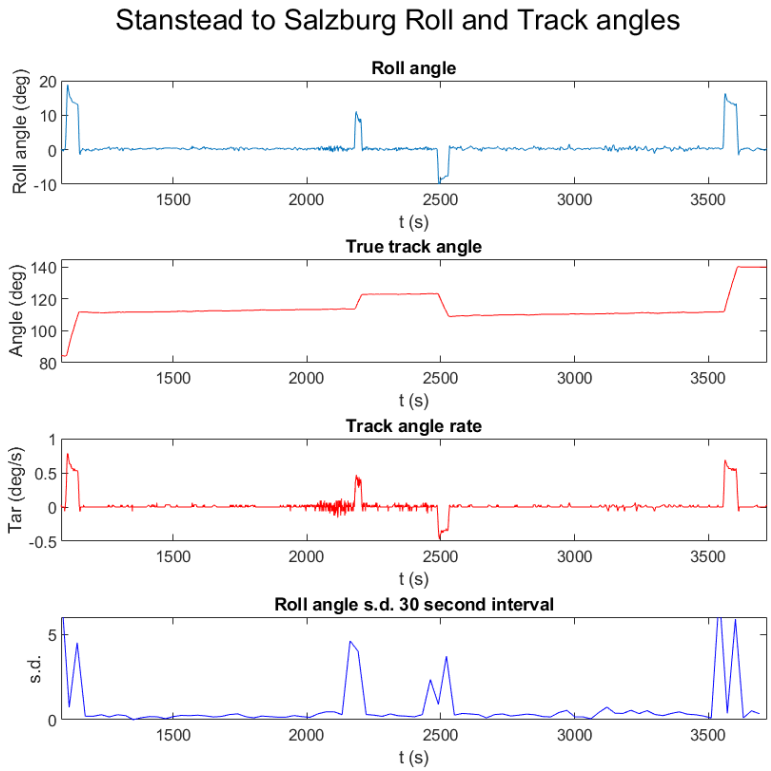



Figure 9: Stanstead to Salzburg Roll and Track angles in the cruise phase with 30 seconds standard deviation intervals

**Modelled flight data**

### 7.1.4 Roll angle data frequency

As the BDS5,0 report averages an output every 1.5 seconds, across a 1000 second interval 667 data points are expected. The frequency of the report was hypothesised to affect the accuracy of the turbulence detection using standard deviation. In Figure 21, the VRSD and RASD were plotted at different average output frequencies: 1000 data points correlated to an output every 1 second, 667 data points to an output every 1.5 seconds and 100 data points to an output every 10 seconds.

To simulate the random outputs, random numbers chosen using a MATLAB code, sorted in ascending order and then indexed against times of Vertical rate and Roll angle and then plotted. This means every time the code is run, a slightly different collection of between 660 and 683 data points are plotted, due to the nature of the randomiser being able to choose the same value multiple times. Similarly for 100 data points, it is between 97 and 103 data points that are plotted.

Figure 10 shows the standard deviation graphs for 1000 data points versus 667 data points are not identical, but the trends are similar enough to conclude that the BDS5,0 does not report too infrequently on average to be determined. However, in real flight data, the BDS5,0 goes through periods of time over 30 seconds where no reports are outputted, which then do lead to entirely missed Roll angle variations and very inaccurate standard deviations calculated. The 100 data point standard deviation graphs confirm this and even indicate turbulence using the $\Delta\sigma_{V_s}$ threshold at different times to the 1000 data point standard deviation graphs. It has been visually determined that where there are fewer than 400 data points per 1000 seconds, the standard deviation trends vary significantly, and the turbulence indications can no longer be considered accurate.

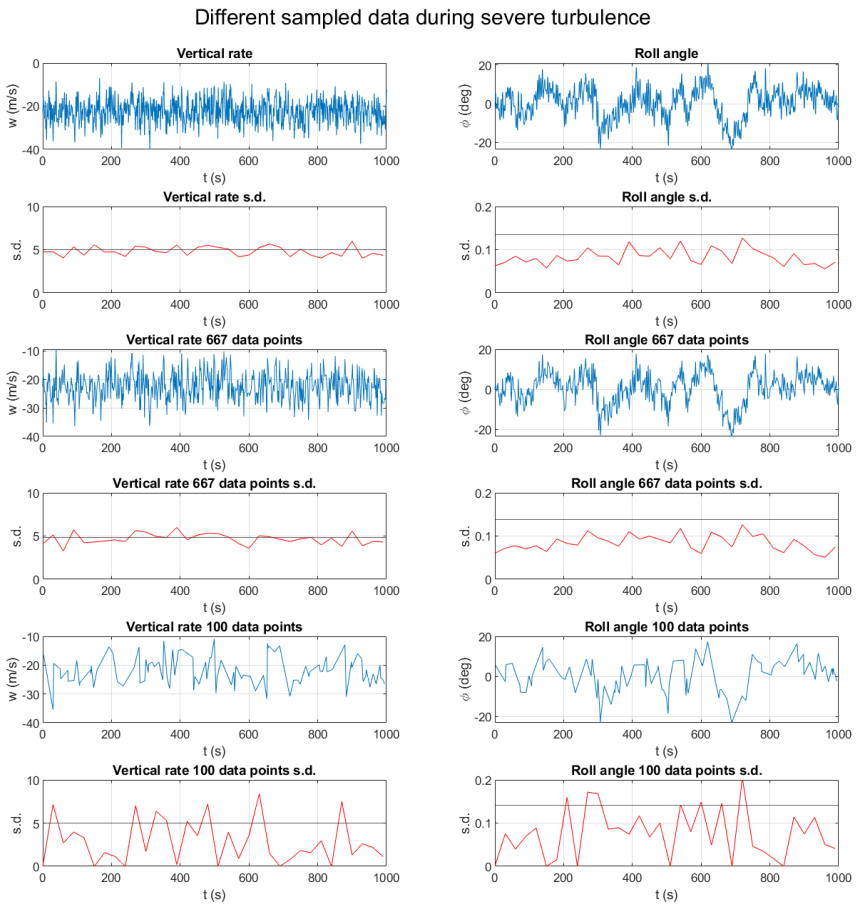



Figure 10: Uniform and sampled Vertical rate and Roll angle data from RCAM model with Severe turbulence

### 7.2 Different turbulence intensities

Figure 11 shows the Vertical rate and Roll angle responses and standard deviations to different intensities of turbulence. The trends between moderate and severe turbulence are similar, but the Y-Axis are the most significant difference. As turbulence intensity increased, the Roll angle range increased as well as the standard deviation, suggesting that there is a correlation between turbulence intensity and standard deviation.

The Roll angle response range is very large: ±10 °s for moderate turbulence and ±20 °s for severe turbulence. This is much larger than any real aircraft would have, as flight control systems include an auto-pilot system, ride control and the implementation of load alleviation during turbulence. This would reduce the response from turbulence, preventing such extreme differences and meaning that the range would be smaller. Between 2550 and 3550 seconds for the Stanstead to Salzburg, the Track angle rate is close to zero, suggesting no aircraft manoeuvres are being conducted. In this time interval of 1000 seconds, the RASD of 30 second intervals are between 0.05 and 0.73. 0.73 is a much higher standard deviation than the RCAM models RASD peak of 0.12. As it stands, the RCAM's Roll angle response is not an accurate replication of real aircraft's Roll angle response to turbulence.

The peak range of the Vertical rate for the Stanstead to Salzburg flight 21.13 m/s, and for severe turbulence, the Vertical rate of the RCAM model varies mostly within a ±10 m/s range about the trimmed value of -22.5m/s. This suggests that as the barometric measurement is unfiltered, the same as the RCAM models output, the RCAM model's response is realistic and verifies the model. The Stanstead to Salzburg aircrafts VRSD peaked at 4.47, across 30 second intervals. The VRSD of the RCAM model

when subjected to severe turbulence is within the range of 4 and 6, further supporting that across at least one 30 seconds interval, the Stanstead to Salzburg flight was subjected to severe turbulence. It can also be seen that the VRSD across the entire time period for the RCAM model lie are also approximately the median values of 30 second standard deviation intervals. Applying these VRSD values to real flight data to determine the intensity of turbulence could be applicable.

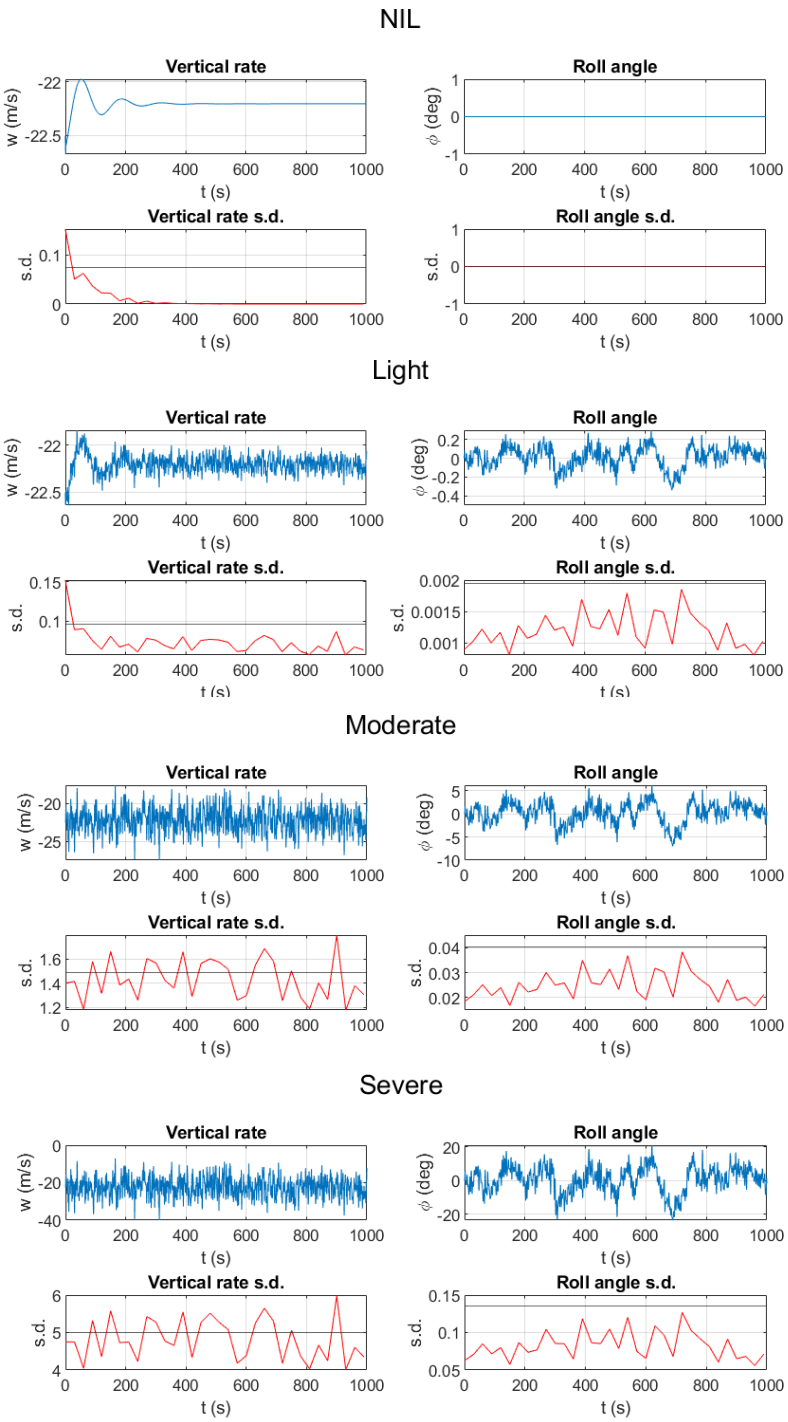



Figure 11: Vertical rate and Roll angle responses from RCAM model to different intensities of turbulence

Table 7: The standard deviations of continuous different turbulence intensities (3 s.f.)

| Intensity | Vertical rate s.d. | Roll angle s.d. |
|---|---|---|
| NIL | 0.0739 | 0.00 |
| Light | 0.0960 | 0.00190 |
| Moderate | 1.49 | 0.0402 |
| Severe | 4.99 | 0.135 |

### 7.3 100 seconds of turbulence

Figure 12 shows the RCAM model being subjected to 100 seconds of different intensities of turbulence, at the 500 second mark. Turbulence can last a short or long period of time, but as the $\Delta\sigma_{V_s}$ threshold equation includes the VRSD of the whole flight, this would significantly vary with the length of time the aircraft is subjected to turbulence. An aircraft subjected to severe turbulence for a long period of time, would have a lower standard deviation than an aircraft subjected to severe turbulence for a short period of time.

Table 8: The standard deviations of 100 seconds and continuous different turbulence intensities (3 s.f.)

| Intensity | Vertical rate s.d. 100 seconds | Vertical rate s.d. continuous | Roll angle s.d. 100 seconds | Roll angle s.d. continuous |
|---|---|---|---|---|
| NIL | 0.0739 | 0.0739 | 0.00 | 0.00 |
| Light | 0.0698 | 0.0960 | 0.00190 | 0.0111 |
| Moderate | 1.88 | 1.49 | 0.0402 | 0.248 |
| Severe | 12.1 | 4.99 | 0.135 | 0.670 |

The VRSD with 100 seconds of turbulence for the severe intensity was higher, but the light and moderate intensities had a smaller difference. The RASD for the continuous turbulence were between 5 and 6 times the RASD with 100 seconds of turbulence. Comparing to real flight data, RASD across the entire flight, for 4 different flights, was in a range of 0.075 to 0.36 (0.0753, 0.1541, 0.1934, 0.3551). This verifies that the RCAM model's RASD occur in a similar range, but that the time turbulence is present relative to the length of the cruise phase of the flight has a significant impact.

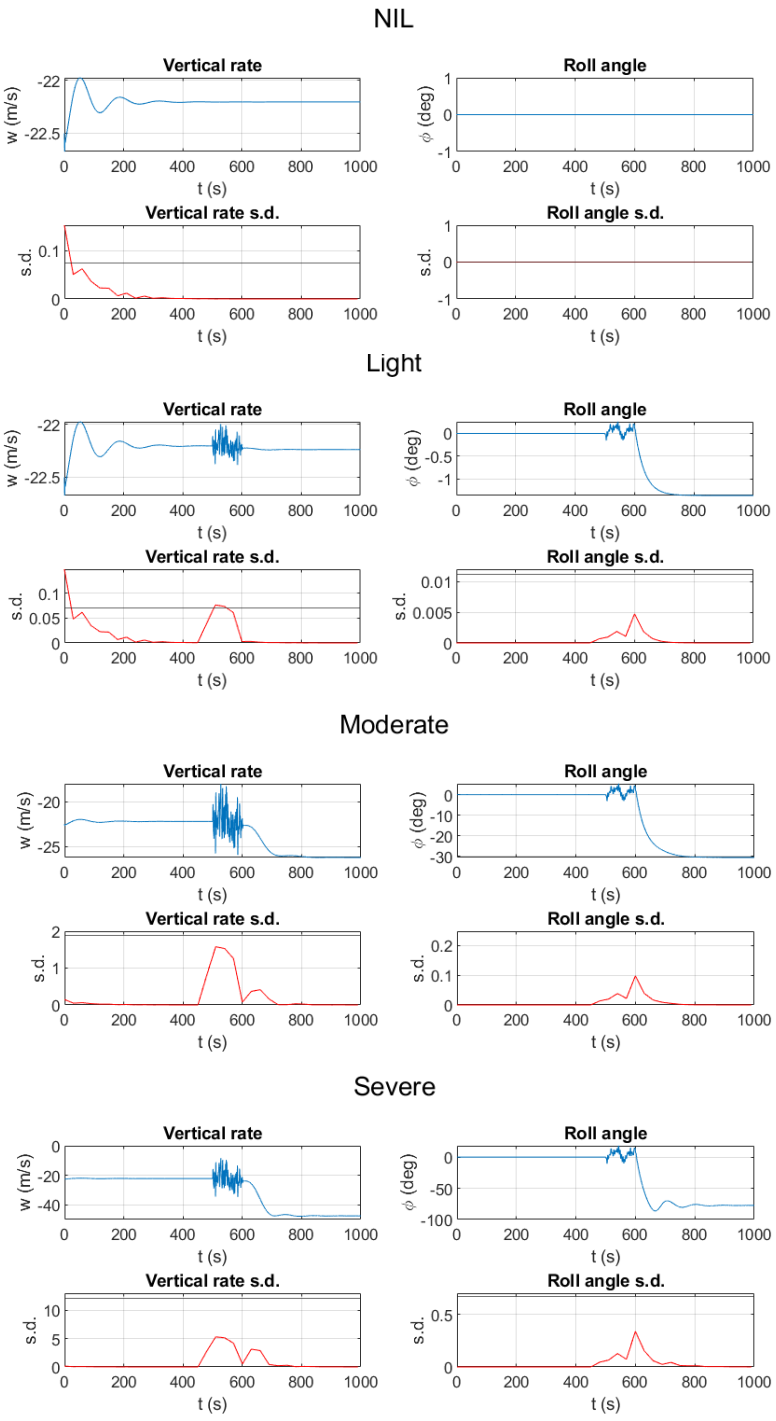



Figure 12: Vertical rate and Roll angle responses from RCAM model to 100 seconds of different intensities of turbulence

### 7.4 Applying RCAM findings to real data discussion

There is no data in the Mode-S report or ADSB report that give the pilot's control inputs. With this data, it could be possible to separate the intentional changes in Roll angle of the aircraft, against the Roll angle changing as a response to flying through turbulence. Flight Data Monitoring (FDM) is a collection of data that is taken from individual airlines aircraft while they are flying and sent wirelessly to the

respective airlines Head Quarters. This means there is no centralised system that can currently access the pilot's inputs, but it stands to reason that this data is sent at an appropriate frequency to map it live against Mode-S Roll angle data.

The irregular frequency in the BDS5,0 plot, and the fact there are over twice as many BDS6,0 data points (worked out from the frequency of reporting), means the Roll angle data is much less accurate than the Vertical rate data used previously by Sun and Olive to predict turbulence. Mode-S EHS data being reported reliably only in European airspace also prevents a worldwide mapping, as it hasn't been widely adopted across different parts of the world. ADSB data also drops out over middle Eastern airspace and part of the Atlantic Ocean. This is speculated to be due to a potential lack of transponders across these areas, or political decisions.

## 8. CONCLUSION

A number of conclusions can be drawn about the suitability of Vertical rate, Roll angle, and BDS reports to indicate turbulence, and the accuracy of the RCAM model's response to turbulence compared to real flights.

Vertical rate data has been verified to show turbulence; however, more accurate verification is needed than rather subjective recounts of turbulence, and across a much larger data set. As it stands, Roll angle data from the Mode-S BDS5,0 report is not currently suitable to map a live indication of turbulence. Roll angle as a parameter to accurately indicate turbulence has not been confirmed. Correlation between turbulence intensities and Roll angle standard deviation have been shown with the RCAM model which has no auto-pilot responses. However, the RCAM model's Roll angle response to severe turbulence ranges of ±20°. This is not representative of a real flights' Roll angle response to turbulence; without separating pilot and auto-pilots control surface inputs, an aircraft's true Roll angle response to turbulence is unknown. Standard deviation taken across time intervals of different lengths do not consistently report the same presence of turbulence at the same times, so more testing is needed to define an accurate time interval for standard deviation to indicate turbulence.

It has been determined that Mode-S data can be decoded and used in a time frame of under 3 minutes. The current mapping of turbulence would be improved by the use of Mode-S data, especially the frequency at which forecasts can be outputted. However, BDS reports are outputted at non-uniform frequencies and on occasion not for time periods of over 5 minutes. This means it has not currently been decided if Mode-S is a reliable data source to provide up-to-date data within the required 3 minute time frame.

As climate change threatens to increase the presence of clear air turbulence, developments in accurate mapping and forecasting will prove to be increasingly more useful. As the adoption of extended Mode-S reporting continues, the BDS4,5 report where pilots are able to report turbulence could be used, however, this is not an automatic report like the Mode-S EHS data, and the likelihood of commercial airlines investing in this technology to install it into their new aircraft is unknown.

## 9. FUTURE WORK

Further research into separating the pilot control inputs from the outputted Mode-S data would be useful to be able to see the aircraft's true response to turbulence. Knowledge of how flight control systems that include ride control affect the aircraft's responses to turbulence would similarly further verify whether the Mode-S outputted data is an accurate representation of the aircraft's response to turbulence. It is unknown how different aircraft implement load alleviation during turbulence, and how it would affect the aircraft's response to turbulence. Further research into the aircraft's response during turbulence would allow for more informative use of outputted Mode-S parameters.

Determining standard deviation thresholds for different intensities of turbulence, and for different lengths of time during the flight where turbulence is present, would allow for discrimination between intensities using a numeric scale and remove the current subjective nature of turbulence reporting. Removing periods of time where there are few data points from the analysis would also improve accuracy.

Live mapping has been shown to be possible within a 3 minute time scale, and the time complexity of decoding Mode-S data. As the use of BDS4,5 reporting is implemented as the technology becomes more widely installed in aircraft, it can be used to verify developing automatic turbulence forecasting techniques, and even the accuracy of other forecasting devices such as weather satellites and radiosondes. Contact with Mode-S transponders is often lost when the aircraft is at a low altitude, such as during the take-off and landing phases of flight. It is at these phases of flight, that turbulence can occur with higher intensities, so further research into how effective Mode-S reporting can be at different phases of flight would allow more understanding of how turbulence affects aircraft in different positions.

## 10. ACKNOWLEDGEMENTS

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## 11. REFERENCES

[1] J. Sun and X. Olive, *Detecting and Measuring Turbulence from Mode-S Surveillance Downlink Data,* ICRAT, 2020.

[2] Met Office, *What is a WAFC?,* [Online]. Available: https://www.metoffice.gov.uk/services/transport/aviation/regulated/mo-wafc. [Accessed 9 May 2021].

[3] Trig Avionics, *MODE-S,* [Online]. Available: https://www.trig-avionics.com/knowledge-bank/transponders/Mode-S/. [Accessed 9 May 2021].

[4] SKYbrary, *Mode-S,* 14 January 2018. [Online]. Available: https://www.skybrary.aero/index.php/Mode_S. [Accessed 9 May 2021].

[5] M. Paydar, *SSR Mode-S Coordination Issues,* ICAO, 2010.

[6] E. Chang and e. al, *The Story of Mode-S,* Lincoln Laboratory, 2000.

[7] SKYbrary, *Transponder Requirements for Access to EU Airspace,* 12 June 2017. [Online]. Available: https://www.skybrary.aero/index.php/Transponder_Requirements_for_Access_to_EU_Airspace. [Accessed 9 May 2021].

[8] Met Office, *Guidelines for interpreting World Area Forecast Centre Significant Weather forecasts,* [Online]. Available: https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/services/transport/aviation/ga/sigwx-interpretation-guide.pdf. [Accessed 9 May 2021].

[9] J. M. Kopec and e. al, *Mode-S and ADSB as a Source of Clear-Air Turbulence Measurements,* 15 November 2015. [Online]. Available: https://meetingorganizer.copernicus.org/EGU2016/EGU2016-481.pdf. [Accessed 9 May 2021].

[10] Federal Aviation Administration, *Meteorological Quality Control for the Calculation of Altimetry System Error,* 19 October 2017. [Online]. Available: https://www.faa.gov/air_traffic/separation_standards/rvsm/documents/ASE/3.7_MET_QC_for_the_Calculation_of_ASE.pdf. [Accessed 5 May 2019].

[11] P. D. Williams, *Increased Light, Moderate, and Severe Clear-Air Turbulence in Response to Climate Change*, University of Reading, 2017.

[12] The COMET® Program, *Forecasting Clear Air Turbulence for Aviation*, 1 March 2016. [Online]. Available: https://www.meted.ucar.edu/avn_int/turbulence/navmenu.php?tab=1&page=3-0-0&type=flash. [Accessed 7 April 2021].

[13] A. K. Mirza and e. al, *Comparison of aircraft-derived observations with in situ research aircraft measurements*, Royal Meteorological Society, 2016.

[14] B. Strajnar, *Validation of Mode-S Meteorological Routine Air Report aircraft observations*, Journal of Geophysical Research Atmospheres, 2012.

[15] J. Sun, *The 1090 Megahertz Riddle: A Guide to Decoding Mode-S and ADSB Signals*, TU Delft OPEN Publishing, 2021.

[16] A. K. Mirza and e. al., *Comparison of aircraft-derived observations with in situ research aircraft measurements*, Royal Meteorological Society, 2016.

[17] M. Kitchen and E. Stone, *Presentation slides*, Met Office, 2017.

[18] Aerospace Technology, *BAE 146*, [Online]. Available: https://www.aerospace-technology.com/projects/bae146/. [Accessed 9 May 2021].

[19] Federal Aviation Administration, *Aeronautical Information Manual*, Federal Aviation Administration, 2020.

[20] J. Sun, H. Vû, X. Olive and J. M. Hoekstra, *Mode-S Transponder Comm-B Capabilities in Current Operational Aircraft*, OpenSky Symposium, 2020.

[21] International Civil Aviation Organization, *Annex 3 to the Convention on International Civil Aviation*, International Civil Aviation Organization, 2010.

[22] GARTEUR, *Research Civil Aircraft Model (RCAM)*, GARTEUR Open, 1995.

[23] Boeing, *757_passenger*, 2008. [Online]. Available: https://www.boeing.com/resources/boeingdotcom/company/about_bca/startup/pdf/historical/757_passenger.pdf. [Accessed 9 May 2021].

[24] SKYbrary, *ICAO Wake Turbulence Category*, 10 December 2020. [Online]. Available: https://www.skybrary.aero/index.php/ICAO_Wake_Turbulence_Category. [Accessed 9 May 2021].

[25] S. Buryev, *Image*, 1 March 2020. [Online]. https://threader.app/thread/1234162253273890817. [Accessed 9 May 2021].

[26] T. M. I. Hakim and O. Arifianto, *Implementation of Dryden Continuous Turbulence Model into Simulinkfor LSA-02 Flight Test Simulation*, Journal of Physics Conference Series, 2018.

[27] MathWorks, *Dryden Wind Turbulence Model (Continuous)*, 2006. [Online]. Available: https://uk.mathworks.com/help/aeroblks/drydenwindturbulencemodelcontinuous.html. [Accessed 9 May 2021].

[28] OpenSky Network, *The OpenSky Network*, 2021. [Online]. Available: https://opensky-network.org/about/about-us. [Accessed 9 May 2021].

[29] OpenSky Network, *A Quick Guide To OpenSky's Impala Shell*, 2021. [Online]. Available: https://opensky-network.org/data/impala. [Accessed 9 May 2021].

[30] J. Sun, *pyopensky 1.4*, pypi, 17 December 2020. [Online]. Available: https://pypi.org/project/pyopensky/. [Accessed 9 May 2021].

[31] J. Sun, *pyModeS 2.9*, pypi, 23 January 2021. [Online]. Available: https://pypi.org/project/pyModeS/. [Accessed 9 May 2021].

[32] J. Sun, *openap 1.1*, pypi, 9 April 2021. [Online]. Available: https://pypi.org/project/openap/. [Accessed 9 May 2021].

[33] SKYbrary, *BOEING 737-800*, 31 July 2016. [Online]. Available: https://www.skybrary.aero/index.php/B738. [Accessed 9 May 2021].

# 12. APPENDICES

## Appendix 1

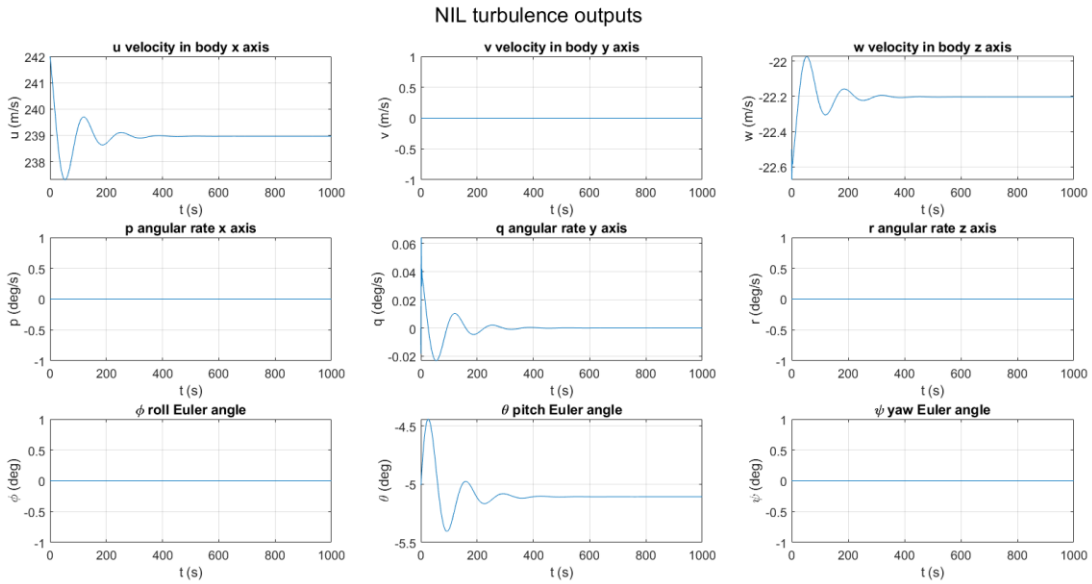



Figure 13: Turbulence Outputs from RCAM model with NIL turbulence

## Appendix 2

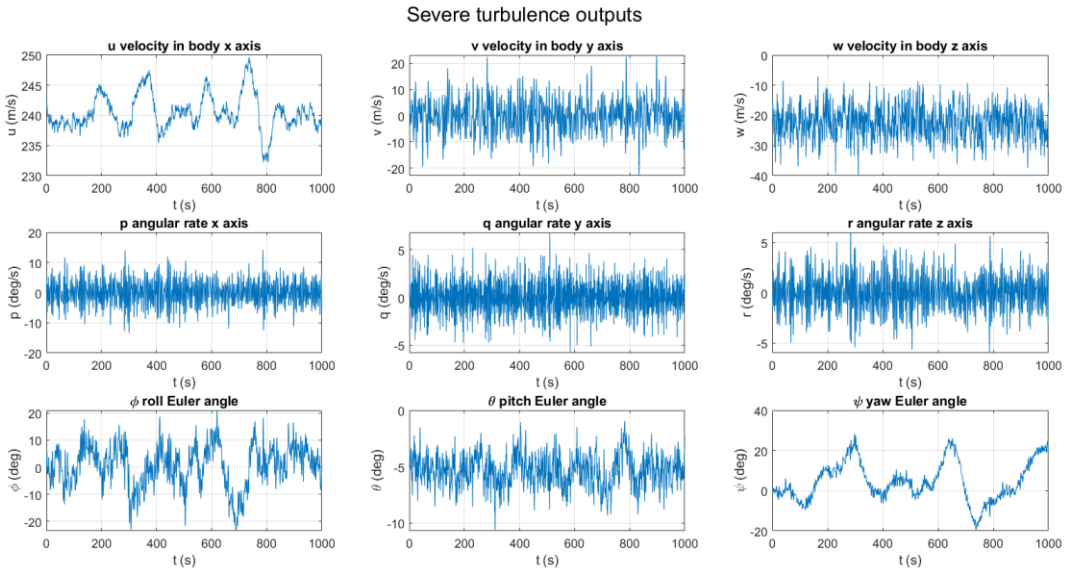



Figure 14: Turbulence Outputs from RCAM model with Severe turbulence

## Appendix 3

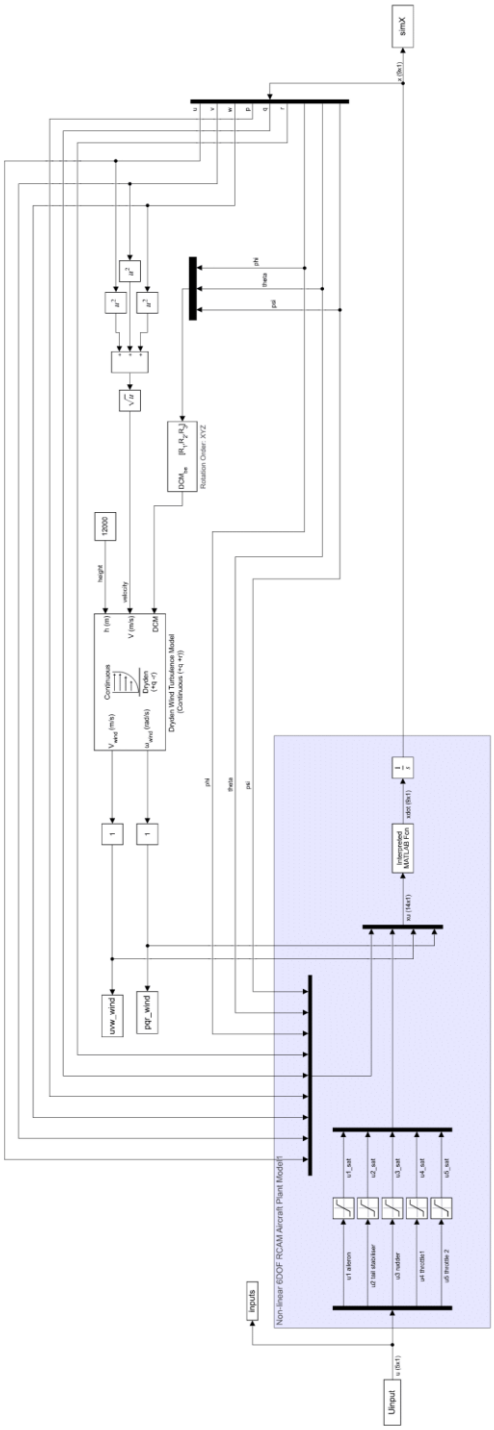



Figure 15: RCAM Simulink model with Dryden turbulence block

## Appendix 4

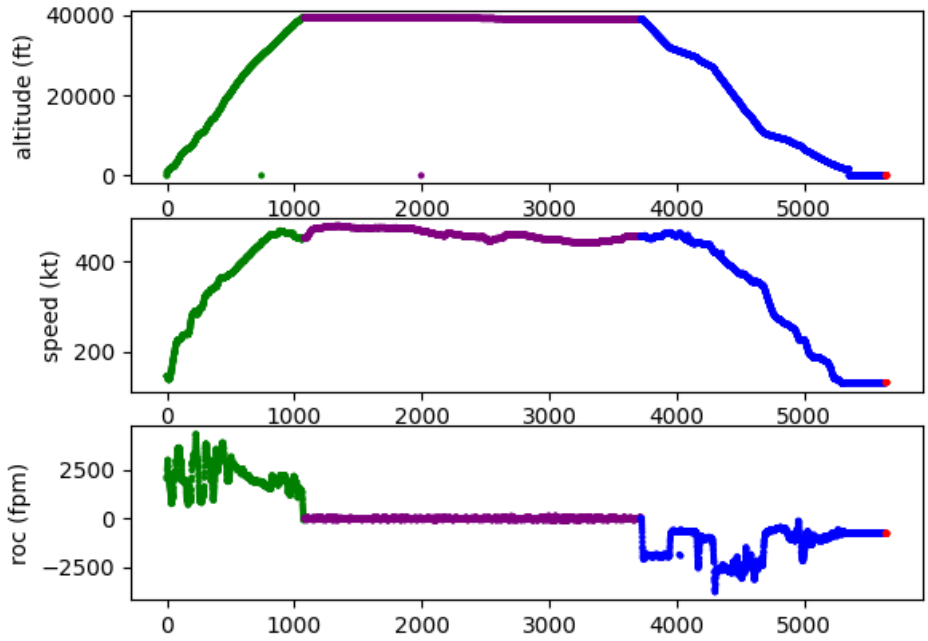



Figure 16: Graphical representation of different phases of flights, with the purple colour representing the cruise phase.

## Appendix 5

Table 9: Calculating reporting frequency of ADSB and BDS reports.

| Flight | ADSB | | BDS5,0 | | BDS6,0 | |
|---|---|---|---|---|---|---|
| | Data points | Frequency (s) | Data points | Frequency (s) | Data points | Frequency (s) |
| STNSLZ | 2640 | 1.00 | 1604 | 1.65 | 4348 | 0.61 |
| LHRTLV | 4720 | 1.00 | 3395 | 1.43 | 10511 | 0.45 |
| AMSPOZ | 1800 | 1.00 | 1060 | 1.70 | 2060 | 0.87 |
| BEGLHR | 6270 | 1.00 | 3790 | 1.65 | 6587 | 0.95 |
| LHRDUS | 660 | 1.00 | 586 | 1.13 | 1040 | 0.63 |
| Average | - | 1.00 | - | 1.50 | - | 0.70 |